

KALUMBURU, Australia

WMO: 941000

Lat: 14.2964S

Lon: 126.6453E

Elev: 24

StdP: 101.04

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.0	11.5	-4.5	2.6	25.9	-2.4	3.1	26.7	8.7	29.4	6.9	28.3	0.7	200	0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	10.6	37.7	23.8	36.8	23.7	35.9	23.9	28.0	33.4	27.3	32.9	26.7	32.4	4.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.3	30.9	25.9	21.2	30.5	25.2	20.4	29.7	90.0	33.3	86.6	33.0	84.1	32.5	36.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.6	5.9	7.4	39.1	1.7	1.2	6.1	40.0	5.1	40.7	4.2	41.4	2.9	42.3
			5.9	28.7	1.4	0.6	4.9	29.1	4.1	29.5	3.3	29.8	2.2	30.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	28.9	28.5	28.6	27.7	25.3	23.0	23.0	23.9	27.0	29.4	30.5	30.1
	DBStd	3.20	1.56	1.47	1.36	1.52	2.33	2.75	2.26	1.82	2.08	1.45	1.26	1.73
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	3	0	0	0	0	0	2	1	0	0	0	0	0
	CDD10.0	6261	587	519	576	532	475	389	404	429	509	602	615	623
	CDD18.3	3222	328	286	318	282	217	141	147	171	259	344	365	365
	CDH23.3	37461	3697	3002	3326	2990	2416	1660	1807	2078	3078	4231	4691	4485
(13)	CDH26.7	18378	1590	1216	1493	1557	1222	740	832	998	1654	2299	2518	2260
(14)	Wind	WSAvg	2.2	2.2	2.0	1.8	1.8	2.1	2.2	2.3	2.5	2.7	2.4	2.2
(15) Precipitation	PrecAvg	1264	325	309	219	67	16	1	6	0	3	27	67	196
	PrecMax	2288	811	810	607	360	103	14	121	5	44	154	193	574
	PrecMin	723	59	30	18	0	0	0	0	0	0	4	4	38
	PrecStd	318	166	151	134	78	29	3	21	1	8	31	42	128
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	36.5	36.7	37.0	35.7	34.6	34.5	34.8	37.5	38.8	38.6	39.3
		MCWB	26.6	25.5	25.2	22.5	21.4	21.0	18.8	18.2	20.5	22.0	23.9	25.1
	2%	DB	35.4	35.0	35.4	35.7	34.6	33.2	33.4	33.7	36.1	37.2	37.4	37.5
		MCWB	26.4	26.3	25.7	23.1	21.1	20.0	18.4	17.9	20.6	22.6	24.1	25.5
	5%	DB	34.2	33.8	34.4	34.8	33.6	32.1	32.3	32.8	34.9	36.1	36.3	35.9
		MCWB	26.1	26.0	25.6	23.2	20.9	19.1	18.2	18.2	21.0	22.9	24.4	25.6
	10%	DB	33.0	32.6	33.2	33.8	32.6	30.9	31.2	31.7	33.8	35.0	35.2	34.6
		MCWB	25.8	25.9	25.6	23.2	20.5	18.5	17.8	18.2	20.9	23.2	24.5	25.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	28.6	28.7	28.0	26.0	23.9	22.8	23.3	28.6	27.8	27.8	28.2
		MCDB	34.2	33.1	33.3	33.1	30.3	29.8	28.8	29.9	33.4	34.6	33.1	34.6
	2%	WB	27.3	27.5	27.5	26.7	24.6	22.8	21.7	22.4	24.6	26.0	26.7	27.1
		MCDB	33.1	32.4	33.0	32.6	30.1	29.0	28.6	29.3	31.3	33.2	32.7	33.4
	5%	WB	26.7	26.8	26.7	25.8	23.8	21.7	20.7	21.7	23.8	25.2	26.1	26.5
		MCDB	32.2	31.8	32.4	31.8	29.1	28.4	27.7	28.3	30.4	31.8	32.1	32.7
	10%	WB	26.2	26.2	26.1	25.1	23.0	20.5	19.7	20.9	23.2	24.7	25.6	26.1
		MCDB	31.4	31.0	31.5	30.6	28.1	27.9	27.2	27.6	29.7	30.9	31.5	32.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	7.9	9.5	12.7	15.0	16.2	17.1	17.0	14.9	12.5	10.6	9.5
		MCDBR	10.2	10.3	11.7	14.4	15.7	16.5	17.8	18.0	15.6	13.9	11.9	11.5
		MCWBR	3.5	4.1	4.4	4.9	5.0	5.5	5.8	5.7	4.4	3.6	3.3	3.5
	5% WB	MCDBR	9.2	9.3	10.3	11.3	11.8	14.4	15.3	14.5	13.2	11.6	10.6	10.1
		MCWBR	3.7	4.3	4.7	5.1	4.7	6.1	5.8	5.1	4.1	3.6	3.5	3.6
(40) Clear-Sky Solar Irradiance	taub		0.423	0.414	0.396	0.384	0.372	0.346	0.344	0.359	0.417	0.446	0.447	0.424
	taud		2.464	2.493	2.540	2.522	2.503	2.535	2.523	2.470	2.354	2.302	2.341	2.446
	Ebn at Noon		922	923	920	898	876	884	895	909	887	884	895	920
	Edh at Noon		120	115	107	103	99	92	95	107	127	138	134	121
(44) All-Sky Solar Radiation	RadAvg		5.71	5.80	5.83	5.79	5.38	5.14	5.42	6.13	6.75	6.98	6.90	6.16
	RadStd		0.64	0.85	0.71	0.48	0.33	0.22	0.18	0.15	0.22	0.31	0.27	0.52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page